Andy Ye

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EDUCATION

University of Washington | *B.S. Computer Science, Paul G. Allen School | GPA 3.63* Expected Graduation: Jun 2028
Coursework: Deep Learning, Autonomous Robotics, Systems Programming, Data Structures and Algorithms
Honors (2023): 4th Place Grand Award, Regeneron ISEF (Computational Biology) | 1st Place, Washington State Science Fair

EXPERIENCE

UW RAIVN Lab | *Undergraduate Researcher* Sep 2026 – Present
Mentored by Ainaz Eftekhar, co-evolving adversarial objects and 3D-printable fingertip augments for a fixed robot gripper.

Brown University, ACT Lab | *Robotics Research Intern (NSF REU)* Jun 2026 – Aug 2026
Advised by Nora Ayanian and Akshay Narayan. Worked on multi-robot systems and built infrastructure.

- **RoboHeist** (co-first author): a Boston Dynamics Spot captures a flag from a maze while evading patrol robots seen through Spot's onboard cameras. Built the ROS 2 and Vicon control stack, trained the onboard patrol detector (**0.33 to 0.98 mAP50** after pretrained models detected nothing), and built a web tool for virtual maze construction with camera blocking.
- **Radio-to-Pose** (2nd author, in preparation): robot teams that localize only relative to each other drift together. Created an interface that uses a radio strength map to pin them down, which reduces group drift error by **74%**.

UW Ubiquitous Computing Lab | *Undergraduate Researcher* Oct 2025 – Present
Advised by Richard Li. Improved a diagnostics algorithm for Alzheimer's symptoms with ML, added sensors to wearable devices.

- **BEACON**: a phone test that screens for episodes of cognitive impairment in Alzheimer's and liver disease patients. The test takes ~70 questions, which exhausts users. Built a Gaussian process algorithm that finds the threshold in ~**20 questions**.
- **PeriPulse** (CHI 2027, under review): made bone conduction headphones that track heart rate during exercise. Led studies, augmented PPG sensors to headphones, showed music induced microvibrations at the sensor do not degrade PPG signal.
- **ElectroNome** (UIST 2026, poster): built the signal pipeline and ran user studies for a voice-free smart glasses interface. An electric-field sensor on the frame picks up any limb tapped in rhythm, and the pipeline **locks onto intent within 5 s**.

Allen Institute for AI (Ai2) | *Undergraduate Researcher* Oct 2025 – Apr 2026
Advised by Matt Wallingford on the PRIOR team, working to bridge the sim to real gap for embodied AI navigation.

- **Bristle**: built a pipeline that reconstructs YouTube real estate tours into **20,000+ continuous navigation trajectories** with depth and camera poses. Used VGGT to build point clouds, and QwenVL for endpoint object detection. Ran it at terabyte scale on a SLURM cluster and released the dataset with a 3D viewer. Demo: https://andyye.bio/bristle/

PROJECTS

Born, Not Trained: a game tester written by an LLM vs. one trained by RL | *JAX* Aug – Sep 2026

- Game studios train reinforcement learning agents to find bugs. Built a framework where an LLM writes and evolves a playtester for games instead, and with zero training it found **91% of planted bugs vs. 54%** for tuned RL at equal compute. Accepted, NeurIPS 2026 AutoMLR workshop.

DAWG'D: attacking vision based web agents | *PyTorch, SLURM* Mar – Jun 2026

- Web agents that act entirely from screenshots can have their inputs tampered by a malicious page. Built an adversarial page editor, which invisibly alters **10%** of a page to make MolmoWeb report believable lies **69%** of the time. Also made the defense against it, a detector catching **99.9%** of tampered pages on unseen sites. Code: github.com/andyye132/DAWG

andyye.bio | *Three.js, GLSL* 2025 – Present

- A portfolio inspired by mobile games! Features an explorable 3D voxel island, voxel engine, my cat, and water shaders.

PUBLICATIONS

- **(Accepted, UIST 2026 poster)** *ElectroNome: Limb-Agnostic Synchronous Gestures via Electric-Field Sensing.*
  Cynthia Dong, **Andy Ye**, Jared Hwang, Zachary Englhardt, Jerry Cao, Juyoung Lee, Shwetak Patel, Richard Li
- **(Accepted, NeurIPS 2026 AutoMLR Workshop)** *Born, Not Trained: Zero-Training Programs Out-Explore Reinforcement Learning for Game Testing at Equal Budget.* **Andy Ye**
- **(In review, CHI 2027)** *PeriPulse: Open-Ear Headphones for Robust Heart Rate Sensing During High-Intensity Exercise.*
  Cynthia Dong, Liam Bakar, **Andy Ye**, Noel Muthuplakal, Zimo Chen, Kurtis Heimerl, Shwetak Patel, Richard Li
- **(In preparation, ECC 2027)** *Radio-to-Pose Information for Gauge-Drift Correction in Cooperative Localization.*
  Aditya Iyer, **Andy Ye**, Nora Ayanian, Akshay Narayan
- **(In preparation, RA-L)** *RoboHeist: Calibration-Aware Active Evasion as a Two-Sided Information Game.*
  Aditya Iyer*, **Andy Ye***, Nora Ayanian (*equal contribution)

SKILLS

Languages: Python, C++, Java, C#, JavaScript, TypeScript, GLSL
ML and robotics: PyTorch, JAX, YOLO, scikit-learn, GPyTorch, ROS 2, MuJoCo, Spot SDK, Vicon
Systems and web: Linux, SLURM, Git, Flask, React, Three.js, Unity, Efficiently using AI coding agents in parallel